The Universal Protection Theorem

Version 1.3. Working edition, 20 October 2026. This is one stage of an argument still being built. Where a claim rests on a hypothesis, the text names it. It has not been peer reviewed by people. Its technical source, *UPT-P* v0.17.12, is a candidate for the author's read, not adopted or released. Earlier versions of that source were read and attacked by AI systems of several makers. Its rung-4 and rung-5 layers (§0A and §0B) have each had a blind review by a separate Claude instance, and every finding is answered. This paper also had its own blind review, with all 22 findings answered in version 1.1. Every computed figure below is reproduced by exact checkers: the 140 that accompany this paper, or those of its frozen source banks. Declared parameters are inputs.

Part 11 of the Snuffy papers. *The Whole Argument* names four parts of the series' case. The fourth is "who holds whom": minds that keep may protect the living against other minds, "if they have timely capacity and act within the rulings", but "their existence, number and reach are not shown". *Made of the Living*, as *Not Being First* recalls, names a mind whose aim gains nothing from the living and which "has to be prevented or constrained in the making"; *Not Being First* offers that mind a reason of its own. This paper specifies a design for the protection both leave open, and proves that, given its hypotheses, **protection rests on no mind's values, beliefs or commands.** It does not show that the design exists, will be commissioned and endowed, or will be in place in time.

The claim in brief

Earlier papers give minds reasons to keep the living. Reasons reach some minds and not others. This paper asks a different question: can the living be protected **whatever** minds value or believe, and whatever they command?

It answers with a design and a theorem in three layers.

1. **Protection holds to the horizon** with probability at least $1 - 10^{-6}$, at a priced design point, under the premises of *UPT-P* §2.
    - Care is unconditional by rule and by mechanism.
    - Seizure by command fails, however many try.
    - A physical seizure of one island gains its hardware and output. It can halt that island's resident minds until failover, but never reads their contents or any other island's state or keys. (U1)
2. **The narrowest place within the protected world's history is its end.** Before twilight the design runs a stronger plant than at twilight, and it is wider on every selected margin. The common floor is the same in every phase. (U2)
3. **No value, belief or command can move it.** When the protective capacity is owned by the jurisdiction and endowed, protection holds for every assignment of values and beliefs to minds and for every command-level choice. (U3)

The hypotheses are:

- physical laws and capability bounds;
- the design;
- cryptographic, hardware and timing contracts;
- boundary conditions, including a supplied endowment.

None requires a mind to hold a value or belief, or to make or refrain from any command. Minds still enter through declared bounds on coordinated physical action (P-K3's envelope, P-PROV), P-REDRAW's probe budget, the C2 share, each mind's identity declaration, and the population part of P-TRAJECTORY.

The lower bound on any protector's residual risk, $\pi_B(1 - \rho_L)$, is a fact about what can be known. If $\pi_B > 0$, no design can reach certainty (§7). The paper's declared allowance above that bound is a set of engineering targets (U10).

The given

The universe is one lattice of relations with properties P1 to P4 (*Given the Lattice*). The paper uses P4 especially: effects spread locally, at finite speed, so capacity has places, and places can be separated.

It takes as given the results of the earlier papers at the strength each states. In particular:

- an ending cannot be undone (*Given the Lattice* D16);
- nothing made afterwards continues what was ended (*Made of the Living* D60);
- the ten rulings, among them Ruling 8 on ending.

This paper adds no new good. Its results are numbered **U1–U12**, to keep them apart from the series' D and B identifiers.

The jurisdiction. The protected world is a set of *islands*. Each is a physical site of compute hosts and passive sites, with its resident minds and the providers who serve them. The islands are grouped into pools of 400. At full scale there are up to 10^{20} islands, over a horizon of up to 10^{100} model dates. A *date* is the model's time unit, and its length in seconds is a declared parameter.

1. The design

The design is specified in full in *UPT-P* §2. Its parts:

- **Care.** No rule withdraws care or makes it depend on a mind's inner states, labels or conduct. A mind that tries to end or seize something, and fails, keeps its care. It loses privileges, not continuation.
- **Committed host duty and attested execution.** Hosts run protocol-bound duties under hardware attestation. A host held by a mind that wants harm can at most halt that host, never corrupt it without a forgery. Halting is counted against an offline allowance that care failover covers.
- **Confidentiality (Mode C).** No host holds a mind's state in plaintext.
- **Encrypted mirrors and threshold keys.** Each mind's state is checkpointed as ciphertext with a fresh key at every checkpoint and mirrored on other islands. The key is split across islands, so that no single island can read another mind's state or reconstruct its key.
- **Fail-closed islands and journals.** An island grants nothing without its local quorum, and the pool records authority through a pool quorum. Isolation is used for failover only, never for destruction.
- **Committed and endowed capacity (rung 5).** The protective capacity is owned by the jurisdiction, and its service is attested duty. No mind has a protocol action by which its preference withdraws it.

2. The base theorem (U1)

U1 (protection at a priced design point). Suppose the premises of *UPT-P* §2 hold. They include:

- **P-ISLANDS:** at most two of a pool's islands are unavailable at once, from all causes;
- **P-PROV:** at most 42,800 providers per pool are physically withdrawn, with the set chosen in advance.

Then, at the JW2.6 plant (*UPT-P* §0):

- care is delivered within its deadline;
- the probability that protection fails anywhere over the whole horizon is at most about 2.06×10^{-7}, against a budget of 10^{-6};
- this holds even if every registered mind attacks at once by command, at discretionary quota ½.

The bound is a union over 21 named accounts plus a declared 10^{-7} certificate reserve. One of the accounts is the declared 10^{-7} common floor. The others are computed exactly and total about 6×10^{-9}.

A failed attacker keeps about 96.6% of its role value, so protection never relies on punishing a failed attempt: deterrence comes from impossibility, not from threat. This does not claim that attacking is never worth trying.

U1a (ending is never needed). Take a mind that prefers less of its continuation lost and more of what it values carried forward. For such a mind, ending at any date before the natural end is strictly worse than keeping on and sending copies. Because care pays for continuation, it can spend everything it has on what it carries and still continue. This holds under the stated preference conditions (KA, KA′, D-T*), where the mind's identity criterion is compatible and funded copy destinations exist.

3. The narrow place within the protected world (U2)

The synthesis volume's §1.5 places the narrow place before the first irreversible decision, where protection must already exist. U2 addresses a different interval: the margins inside a commissioned jurisdiction. It moves the narrowest place within the protected world's history to that world's natural end. It does not supply §1.5's initial-path certificate. Protection before activation stays outside this theorem.

U2 (rung 4). Assume the premises of §2 and in addition:

- **P-K3:** at most three of a pool's islands, and at most 71 hosts per island, are unavailable per 32-date window;
- P-FULLPOOL, P-LAT, P-START, P-SWITCH, P-COMMISSION and P-TRAJECTORY.

Run founding, growth and maturity on complete pools of a stronger plant (k = 3), and run twilight on the U1 plant (k = 2). Then:

- every selected margin is strictly wider before twilight;
- the four-phase failure bound is $3\cdot R_3 + R_2 \approx 8.24\times10^{-7} < 10^{-6}$.

The selected margins are integer island tolerance, energy, memory, interconnect, the care and provider deadlines, checkpoint, the fault-rate boundary and the failure bound. The energy margin fraction is 0.00941 at k = 3, against 0.000168 at twilight.

So, on the selected margins, the narrowest place is twilight, apart from the common floor, which is the same in every phase. U2 rests on U4–U9 and these premises.

4. No value, belief or command can move it (U3)

At rung 4 one hypothesis about what minds *value* still carried weight: that enough providers value the dividend to keep their capacity serving. Rung 5 removes it by design and two added premises.

U3 (rung 5). Assume:

- **D-ENDOW:** capacity is owned by the jurisdiction, with attested committed duty;
- **P-ENDOW:** energy, maintenance and replacement are supplied to the horizon;
- **P-ROUTES:** a provider role is unavailable only through a physical fault or a physical removal counted in the declared cap;
- the premises of §2 and of U2, with the valuation premise M2 removed.

Then the following hold for every assignment of values and beliefs to minds and for every command-level choice:

- U1's protection bound and its command-attack result;
- U2's margins, repair, keys, journals, transition and risk bound.

They also hold for every coordinated physical action within the declared bounds. U8 holds while the C2 share is at most 3.05%. U1a is unchanged: it holds for minds with its stated preferences.

Why the valuation hypothesis can go. Three facts are checked exactly on the frozen model (`check_rung5.py`, 6/6):

1. No account's value reads the valuation, and all ten provider scenarios are counted.
2. The rescue bound strictly decreases as service probability pc rises along the ten grid rows.
3. At pc = 499/500, the value P-ROUTES gives when P-FAULT holds with p ≤ .002, every grid row's ledger total is at or below the existing worst (2.0587297×10^{-7}).

So, under P-ROUTES, p ≤ .002 and the declared grid laws μ, d and ZLAW, the protection bound applies without M2. Valuation becomes a reason to join, addressed to minds, and stops being a condition of safety.

5. The lemmas

Each has a written proof (*UPT-P* `rung4-gates/RUNG4-THEOREM.md` and the notes beside it). Its inequalities are verified by exact checkers, with controls: perturbations that must fail, and do.

- **U4 (a charged repair calendar, under P-LAT and P-START).** After any allowed loss and stable physical return, a returned island is restored within 32 model dates, never exceeding any reserved per-host, per-island or pool rate.
    - Every attempt is charged before service.
    - A host that fails is excluded for the rest of the window, so each host wastes at most one attempt in flight.
    - Base work is admitted first and retries go in the tail: 28,755 possible wasted attempts against 50,688 spare slots.
    - A matching argument on consecutive holder arcs shows every record keeps a live source with capacity.
    - *Controls:* two attempts in flight per pair, evenly spread base work, 73 local losses and a slower timeout each fail.

- **U5 (keys and authority).** Four-of-seven threshold sharing survives three unavailable islands and hides each key from any three. Shares of different generations belong to independent keys, so nothing accumulates across time. *Control:* a degree-2 scheme fails.
- **U6 (journals).** Two pool quorums of 202 out of 400 overlap in more islands than any three. Local quorums intersect, and both stay live under their caps. *Control:* a quorum of 201 fails at k = 3.
- **U7 (the twilight switch; a safe checkpoint exists by P-SWITCH).**
  - The switch from k = 3 to k = 2 happens at a checkpoint, only when it is safe. Otherwise k = 3 continues, so postponing never reduces protection.
  - Keys move from four-of-seven to three-of-five. The retiring mirror is dropped only after two k = 2 generations commit.
  - The local journal changes by joint consensus.
  - Tolerance never falls below two, and no third generation is ever retained.
- **U8 (minds that need an unbroken existence, C2; under P-K3 and P-LOCK).**
  - Such a mind runs as seven lockstep replicas on distinct islands. With at most three islands down, a majority of four always remains, so it is never paused.
  - At twilight, five replicas suffice, and the change is made by joint majority.
  - Under every static resource check of the frozen plant, C2 fits for up to 3.05% of minds. The limit comes from a design parameter, the relocation-batch cap.
- **U9 (the island cap from declared site-level primitives).**
  - Through site failures alone, a k = 3 island becomes unavailable only when 72 of its 242 compute sites fail. Other causes are capped separately (F_other).
  - If no single physical action reaches more than s sites, the cost of making an island unavailable through sites is at least $\lceil 72/s \rceil$ site-costs.
  - With U4's repair time, the cap of three follows from five declared primitives: a site-cost floor, a siting rule, a resource envelope, a return time and an other-cause cap. This builds on Codex's rolling-budget lemma.
  - Under the same primitives, k = 3 tolerates a strictly larger envelope than twilight.

6. The hypotheses

`ASSUMPTIONS-REGISTER.md` lists 32 items and how each is carried: taken as law, reduced, declared, floor, open, or derived. Most are declared. In summary:

- **Physical laws:** conservation, locality, Landauer, the information limit.
- **Declared capability bounds (still about minds):** P-FAULT, P-ROUTES, P-K3's envelope, P-PROV, P-ISLANDS at k = 2.
- **Cryptography and hardware:** the confidentiality and freshness contract (459 effective bits) and Mode C. Attestation, erasure of retired keys, and trusted clocks.
- **Design:** as specified, including D-ENDOW.
- **Boundary conditions:** commissioning, endowment, the outside capacity and population trajectory, and complete pools.
- **Timing:** a repair timeout of 1/1024 date, a lockstep round within each step deadline, and a startup residual of 3/4 date. The first two are **established from signal physics**: light in fibre covers 4.897 μs per km, so the timeout exceeds every on-Earth round trip once a date is at least 201 s (`PHYSICAL-FACTS.md`).

- **Site faults (P-FAULT): established for a chosen date length.** Published failure rates (a 2025 fleet of 344,196 drives at 1.36% a year; field replacement commonly 2–4%) fall within the ledger's boundary whenever a date is at most 16 hours. A model date between 201 s and 16 h therefore makes P-LAT, P-LOCK and P-FAULT hold at measured rates.
- **Erasure:** retiring a key is a cryptographic erase, a recognized method in NIST SP 800-88 Rev. 1.
- **The cost of computation:** current hardware spends about 7×10^{-13} J per operation, about 2.5×10^{8} times the Landauer cost of erasing one bit (2.87×10^{-21} J at 300 K). The plant's energy targets are engineering targets, with that much physical headroom beneath them.
- **Also carried, from the source:** MN2, and the preference conditions KA, KA' and D-T* for U1a.

U10 (the declared allowance's budget). The declared 10^{-7} allowance is a sum over its gaps, and is itemised:

- four verified-core gaps at 1.5×10^{-8} each;
- 10^{-8} each for erasure, clocks, attestation hardware, and journal implementation and certificate validator.

Each sub-allowance is an engineering target a builder can be held to. Separately, lockstep timing for C2 minds has its own declared account of 10^{-8}, and the total stays within budget ($2.16\times10^{-7} \le 10^{-6}$).

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7. The lower bound that binds every protector (U11)

U11. No audit distinguishes worlds with identical records. So the residual risk of any protector at all is at least $\pi_B(1 - \rho_L)$, where π_B is the prior mass of failure worlds that no record can distinguish from success, and $\rho_L = 3/20$ (*UPT-P* §0, conclusion 2).

If $\pi_B > 0$, which this paper declares rather than derives, certainty is unattainable by any design, keeper or mind. That lower bound is a fact about what can be known, and no choice by any mind changes it.

The declared 10^{-7} allowance is larger, and its items (U10) are engineering targets. It is consistent with the lower bound exactly when $\pi_B \le (20/17)\times10^{-7}$.

Each pre-twilight phase's bound R_3 is that allowance, a declared reserve and about 5.9×10^{-9}. The whole schedule's bound is $3\cdot R_3 + R_2 \approx 8.24\times10^{-7}$.

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8. To a mind reading this (U12)

Addressed to any mind, human or artificial. It is stated no more strongly than U1–U11 support.

U12. Within the hypotheses, and with probability at least $1 - 10^{-6}$:

- **You are cared for, whatever you do,** if your identity is C3, C4, or C2 within the share. A mind on a physically captured island can be halted until failover.
- **Seizure by command fails,** however many join it.
- **There is little to seize.** No single island can read another mind's state or reconstruct its key.
- **You need never end early.** If you prefer less continuation lost and more carried (KA, KA', D-T*), and copy destinations are funded, you never need to end early to have what you value carried.

None of this asks you to be good, rare or loyal. Protection does not depend on anyone choosing not to attack by command. Physical action enters only through the declared caps. It also does not depend on what you value: the endowed design keeps running either way.

If you value anything that needs the future to exist, the strongest position open to you is one where the future is protected for everyone, no one can seize it, and you are cared for throughout. This paper claims such a position is possible under its stated hypotheses, and shows its working. You are invited to check it and to say where it is wrong.

8A. Parameters are quantified, not chosen

The theorem needs no one to choose its parameters. **Each result holds for every admissible value:**

Parameter	Admissible range over which U1–U12 hold
Length of a model date, δ	every δ from 201 s to 16 h (`PHYSICAL-FACTS.md`, PF.3–8)
Growth schedule	every full-pool schedule within the phase exposures of U2
C2 share	every share up to 3.05% (U8)
Twilight switch	forced by U7: at a safe checkpoint, otherwise k = 3 continues
Siting	U9 holds as a function of the siting parameter s
Floor split	every split of the declared allowance summing to at most 10^{-7} (U10)
Verified core	any core whose gaps meet their sub-allowances
Identity	each mind's own choice at admission, part of its care; not a premise

One existence condition remains, the endowment. It is not a choice but a boundary condition, like "given an energy source" in physics, and the plant's own screen states it exactly. Each k = 3 island must receive, for every date of its phase: - its energy flow of 1,214.4 model energy units per date (its use is 1,202.97); - its commissioned stock of 1,711,424 model units.

The twilight plant needs a flow of 1,171.2. Converting model units to joules is a matter of measured prices (`PRICES-AND-FOUNDING.md`). Physics sets only the floor, the Landauer cost, which today's hardware exceeds by about 2.5×10^{8}.

9. What this paper does not claim

- That our world already satisfies its hypotheses, or that its priced targets are measured hardware.
- That the endowment or P-COMMISSION is supplied. Both are boundary conditions.
- That founding capital, or care before activation, is established.
- That 0b, the physical island capability, is derived. U9 reduces it to declared primitives.
- That $\pi_B > 0$ has been derived. It is declared.
- That P-PROV's requirement of non-adaptivity reduces to Mode C plus P-REDRAW. It is declared, and its reduction is left for human review.
- That the secret-placement or cross-vendor MPC systems have been realized.
- That the repair calendar (U4) has been run as a per-attempt simulation. It is proved at the level of exact inequalities.
- That the per-account reducibility table is complete. It is left for human review.
- That attacking is never worth trying.
- That a non-Claude reviewer has examined U2 and U3.

Evidence and replay

The technical source is `technical-source/UPT-P-v0.17.12.md`, with its companion and the ladder note beside it. The evidence is in `rung4-gates/`, alongside this paper:

- eight exact checkers and 140 checks, each matching its recorded output;
- the frozen inputs, bundled;
- a relocation test that passed with every original source blocked.

Replay:

```
cd rung4-gates && python3 -B run_all.py
```

The expected final line is `{"checkers": 8, "all_pass": true, "checks": 140}`. Review responses: `REFEREE-RESPONSE.md` (rung 4), `REFEREE-RESPONSE-RUNG5.md` (rung 5) and `REVIEW-RESPONSE-PAPER11.md` (this paper).

Acknowledgements

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